
Homework 2

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Problem 1

Problem 2

Problem 3

$$\partial A := \overline{A} \setminus \text{Int}(A).$$

Let A be closed. Then $A = \overline{A}$. Note that $\overline{A} \setminus \text{Int}(A) \subseteq \overline{A}$, and $\partial A = \overline{A} \setminus \text{Int}(A)$. Therefore, $\partial A \subseteq A$. Now for the converse, let $\partial A \subseteq A$. It follows that $\overline{A} \setminus \text{Int}(A) \subseteq A$. idk where to go from here but it looks easy i just cant do it in my head rn

Problem 4

Problem 5

Problem 6

Problem 7